

Homework — 1.3.3 Networks — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. **[2]** Any two for 1 mark each: a LAN covers a small geographical area / single site, whereas a WAN covers a large geographical area (cities/countries); a LAN's infrastructure is usually owned by the organisation, whereas a WAN typically uses third-party/leased infrastructure; LANs typically offer higher transmission speeds. (The internet is the largest example of a WAN.)
2. **[2]** A protocol is a set of rules governing how data is transmitted/formatted between devices (1). Because the rules are standard/agreed, hardware and software from different manufacturers interpret the data in the same way and can therefore communicate/interoperate (1).
3. **[3]** Up to 3 from: the browser/computer sends the domain name to a DNS server (1); the DNS server looks up / resolves the domain name into the corresponding IP address (passing the request up a hierarchy of DNS servers if it does not hold the record; results may be cached) (1); the IP address is returned to the browser (1); the browser uses that IP address to send its request to the correct web server (1). Max 3.
4. **[3]** Two advantages (1 each, max 2): security/user accounts/access levels are managed centrally; backups can be performed centrally; files and software are stored/updated in one place; easier to manage as the network grows. One disadvantage (1): the server is a single point of failure; expensive — dedicated server hardware and technical staff needed; the server can become a bottleneck under heavy demand. Must have two advantages and one disadvantage.
5. **[3]** Up to 3 from: the file is split into packets, each with a header containing the source and destination (IP) addresses and a sequence number (1); each packet is routed independently across the network by routers and different packets may take different routes (1); at the destination the packets are reassembled into the correct order using the sequence numbers (1); missing/corrupted packets are detected and retransmitted (1). Max 3.
6. **[1]** A firewall monitors incoming and outgoing traffic and blocks/permits it according to a set of rules (packet filtering), preventing unauthorised access to the network (1).

Part B (6 marks)

7. **[2]** Every foreign key value must match an existing primary key value in the table it references (1). This keeps links between tables consistent — e.g. a record cannot reference a record that does not exist, and a record cannot be deleted while other records still refer to it (prevents orphaned records) (1).
8. **[2]** Any two for 1 mark each: one-way / cannot be reversed to recover the input; deterministic — the same input always produces the same hash; low chance of collisions (different inputs rarely give the same hash); quick to calculate; produces a fixed-length output.
9. **[2]** The linker combines the compiled object code of the program with the object code of the library modules/subroutines it uses (1) and resolves the addresses/references between them so they form a

single working executable (1).

Total: 14 + 6 = 20 marks